VisioSphere — Calculations & Tools

Live configuration, cctv_core.py as of 9 September 2026. Formulas as implemented, with the topic to search for each.

A1. CORE — feeds every detection

Calculation	Formula / rule	RRL search topic

Detection gating	`DET_CONF 0.10`
	`MIN_CONFIDENT_KPTS 5 / 17`
	`POSE_CONF_THRESHOLD 0.45`
		Confidence thresholding in pose estimation; recall–precision trade under occlusion

Tracking	`ByteTrack, two-stage association by detection score` `high 0.15 / low 0.05 / new 0.10 / buffer 90 / match 0.85`	ByteTrack; low-score association; ID switches under rapid shape change
Keypoint smoothing	`x = 0.40*x_raw + 0.60*x_prev` `position frozen if conf < 0.15`	Temporal smoothing of pose keypoints; EMA / low-pass; jitter in per-frame pose
Pixel-aspect correction	`k = (W_src/H_src) / (W_inf/H_inf)` `tan(true) = k * tan(measured)`	Pixel aspect ratio; anisotropic scaling; viewpoint distortion in monocular measurement
Ghost rejection	`torso_len / shoulder_width >= 0.60`	False-positive rejection in pose estimation; anthropometric proportion constraints

A2. FALL + LYING DOWN — shared geometry, one state machine

Calculation	Formula / rule	RRL search topic
Torso angle	`theta = | degrees( atan2(dx, dy) ) |` `shoulder-mid to hip-mid ; 0 = upright`	Trunk inclination angle as fall indicator; body orientation from pose
Bbox aspect ratio	`aspect = h / w` `>1.5 upright , <0.9 fallen`	Fall detection using bounding-box / shape aspect ratio
Height collapse	`height_ratio = bbox_h / max_bbox_h` `reference set on upright frames only`	Silhouette height collapse; adaptive standing-height reference
Body extent ratio	`extent = bbox_h / torso_len`	Scale-invariant fall features; calibration-free anthropometric normalisation
Joint angles	`arccos( (v1.v2) / (|v1||v2|) ) cosine rule` `knee = (hip,knee,ankle) ; hip = (sh_mid,hip,knee)`	Joint angles from 2D pose; seated vs supine posture classification
Standing-bend guard	`(ankle_y - hip_y) / torso_len >= 1.2` `no ankle: torso_inv >= 0.30 AND knee_drop >= 0.60`	Bending vs falling discrimination; ADL false positives
Furniture / floor zones	`cv2.pointPolygonTest(polygon, support_point) >= 0` `support_point = hip midpoint`	Context-aware fall detection; bed / chair region-of-interest priors
Path A (horizontal)	`theta > 65 AND aspect < 0.85 AND not bend`	—
Path B (collapse)	`height_ratio < 0.50 AND aspect < 1.20 AND extent < 2.5`	—
Temporal confirmation	`3 consecutive frames (STUMBLE 4, NORMAL 2)`	Temporal filtering / multi-frame confirmation for false-alarm reduction
Size + edge gates	`bbox_h < 0.08 * 360 = 28.8 px` `within 4 px of frame border`	Minimum subject resolution for pose; truncation and occlusion limits

A3. FALL only — what separates it from lying down

Calculation	Formula / rule	RRL search topic
Peak angular velocity	`rate = (theta_last - theta_min) / (t_last - t_min)` `2.5 s window ; >= 45 deg/s = FALL`	**Trunk angular velocity; distinguishing falls from voluntary lie-down; speed of posture change**
Time-bounded history	`prune samples older than 30 s (not a frame count)`	Frame-rate independence in video analysis

A4. LYING DOWN only — positive-evidence gate (LYING_EVIDENCE=1)

Calculation	Formula / rule	RRL search topic
Sustained confirmation	`hold FALLEN 3.0 s before alerting`	Dwell-time / duration threshold; transient-pose rejection
MANDATORY 1 — legs visible	`both hips AND at least one knee at conf >= 0.45`	Lower-body visibility as a precondition; occlusion limits in posture classification

Calculation	Formula / rule	RRL search topic

MANDATORY 2 — not stacked	`stack = (max(ankle_y | knee_y) - sh_mid_y) / torso_len`
	`must be <= 1.20`
	`standing 2.5-3.0 , bending 1.4-2.0 , seated 1.1-1.8 , lying 0-1.0`
		Vertical body extent normalised by torso length; bending vs lying discrimination

Box not tall	`aspect = h / w <= 1.60`	Bounding-box aspect ratio as posture evidence
ALT A — legs extended	`max( hip_flexion ) >= 140 deg (straighter hip decides)`	Hip flexion angle; supine vs seated classification from 2D pose
ALT B — curled but flat	`aspect < 0.85 AND extent < 2.5 AND torso_angle > 65 deg`	Flat-silhouette / compactness evidence for a body on the ground

Evidence persistence	`>= 60% of frames must pass, min 3 frames`
	`4 consecutive misses cancel`
	`rejection held 20 s`
		Temporal voting / evidence accumulation for false-alarm reduction

A5. PROLONGED FALL

Calculation	Formula / rule	RRL search topic
Milestone escalation	`30 / 120 / 300 / 900 / 1800 / 3600 s` `max 6 alerts ; measured from onset`	**"Long lie"** — time on floor after a fall; alarm fatigue / alert burden

A6. INACTIVITY

Calculation	Formula / rule	RRL search topic
Centroid + EMA	`cx = x / frame_w ; cx = 0.3*new + 0.7*prev`	Trajectory smoothing
Stationarity test	`median( v_i over 2 s ) < 0.015` `v_i = hypot(dcx, dcy) / dt`	Median filtering; robust statistics for noisy trajectories
Immobility threshold	`1200 s ; milestones x1, 2, 4, 8 ; max 4`	**"Long lie"**; immobility detection in elder care; threshold duration
Posture qualifier	`theta > 40 deg -> POSTURE variant` `exempt if aspect > 1.5`	Slumped / abnormal posture detection

A7. AGITATION — rhythm-or-vigour oscillation test

Calculation	Formula / rule	RRL search topic
Hands-on-head	`wrist_y < nose_y AND |wrist_x - nose_x| < 1.0 * shoulder_width`	Hand-to-head gesture; agitation / BPSD markers in dementia
Continuity	`6.0 s window ; keep the LONGEST unbroken run` `break if gap > 1.5 s ; run must span >= 3.0 s`	Segmenting continuous movement episodes; avoiding spurious rate from disjoint samples
De-noising	`3-sample median filter on the wrist series` `raw series retained for the vigour path`	Median filtering of keypoint trajectories; white high-frequency keypoint error
Travel gate	`travel = P95 - P05 of raw wrist y` `must be >= 0.50 * shoulder_width`	Amplitude thresholding in shoulder-widths; percentile-based robust range
Swing detection	`step must exceed 0.35 * shoulder_width` `raw track judged at 1.5x that bar`	Zero-crossing / peak detection for repetitive movement

Signature 1 — RHYTHM	`>= 4 reversals ; rate >= 0.8 /s`
	`CV of inter-reversal gaps <= 0.55` (smoothed)
	`or <= 0.40 (raw) ; mean gap >= 1.75 * dt`
		Coefficient of variation of inter-event intervals; periodicity vs Poisson-random noise; repetition rate in agitated movement (30-90 rpm)

Signature 2 — VIGOUR	`raw travel >= 1.2 * shoulder_width` `AND >= 4 raw reversals`	Amplitude-only detection when rhythm aliases above the sampling rate
Score + sustain	`0.20 + 0.60*osc (both hands) ; 0.15 (one) ; 0.25 (elbows)` `alert if > 0.25 held 15 s, 3 s grace, 300 s cooldown`	Hysteresis in behavioural scoring; sustained-threshold alerting

Validation already done (mc_agitation.py, 12 seeds × 90 s per condition): static hand on head 0/12 false alerts at 3, 4 and 5 fps; slow slide 0/12; pulling at 30 rpm 12/12 detected; 50 rpm 11-12/12; ragged irregular pulling 12/12. Known miss: 70 rpm is 3-6/12 at 3 fps and 12/12 at 5 fps; 90 rpm is 0/12 at 3 fps — a sampling limit at Nyquist, not a threshold problem.

A8. DELIVERY — applies to every alert

Calculation	Formula / rule	RRL search topic
IoU	`inter / (area_a + area_b - inter)` `re-assoc >= 0.35 / 3 s ; dedup >= 0.25 / 180 s`	IoU data association in multi-object tracking
Circuit breaker	`< 30 alerts per camera per rolling hour`	Alarm fatigue; alert burden in clinical monitoring
Night gate	`lum = mean(frame) ; engage < 18 , release > 22`	Low-light degradation of detection; hysteresis (Schmitt) thresholding

B1. TOOLS — AI core, computer vision

Tool	Version	Role
YOLO11m-pose	`20.9 M params`	Person detection + 17 COCO keypoints per person.
OpenVINO	`2026.3`	Runs the model on Intel integrated graphics at FP16. Replaced NVIDIA CUDA.
Ultralytics	`8.4`	Model loading, frame preparation, tracking integration.
ByteTrack	`bundled`	Per-person track IDs. Tuned in `bytetrack_fall.yaml`.
PyTorch	`CPU build`	Ultralytics dependency. Not used for inference.
OpenCV	–	RTSP capture (TCP), resize to 640x360, polygon tests, overlays, clip writing.
NumPy	–	All geometry above: angles, ratios, velocities, median test.

B2. TOOLS — AI core, service layer

Tool	Version	Role
Flask + Flask-CORS	–	MJPEG feed, signed clip/upload endpoints, /status health check.
python-socketio	`client`	Sends alert events to the backend.
python-dotenv	–	Loads .env so every threshold above is tunable without code edits.
hmac + hashlib	`stdlib`	Short-lived signed tokens for stream and clip access.
FFmpeg	`external`	Transcodes clips to H.264 / yuv420p. Required — OpenCV cannot write H.264.
clip_server.py	`systemd unit`	Clip archive on a separate process from detection.

B3. TOOLS — backend and frontend

Tool	Version	Role
Node.js + Express	`22.x / 5.2`	REST API and application server.
Socket.IO	`4.8`	Receives alerts from the AI core; broadcasts to dashboards.
MongoDB + Mongoose	`9.2`	Incidents, residents, staff, assessments, audit logs.
JWT + bcryptjs	`9.0 / 3.0`	Authentication and password hashing.
Firebase Cloud Messaging	`13.10`	Push notifications to staff and guardian devices.
Resend	`6.9`	One-time passcodes for login and password reset.
node-cron / ExcelJS	`4.2 / 4.4`	Scheduled archiving; report and audit export.
Helmet / rate-limit	`8.1 / 8.5`	Security headers and request throttling.
React + Vite	`19.2 / 7.3`	Dashboard framework and build tool.
MUI + Tailwind	`9.0 / 3.4`	Interface components and layout styling.
Recharts / MUI X Charts	`3.8 / 9.0`	Incident statistics and trend charts.
Axios / socket.io-client	`1.13 / 4.8`	REST calls and live alert reception.
jsPDF + html2canvas	`4.2 / 1.4`	Client-side PDF report export.

B4. TOOLS — deployment and hardware

Tool	Version	Role
Dell OptiPlex 7010 Micro	`i5-13500T`	On-premise AI box. Intel UHD 770, 16 GB RAM.
Ubuntu	`24.04`	Operating system on the AI box.
systemd	–	Supervises detection and clip archive under /opt/visiosphere.
Cloudflare Tunnel	`cloudflared`	Exposes the AI box with no public IP.
Render	–	Hosts the Node backend.
Tapo TC65	`RTSP / H.264`	IP camera. Sub-stream feeds inference; RTSP forced to TCP.

Tool	Version	Role
Measured performance	`12.18 fps`	81.8 ms per frame. ~12 W idle, ~40 W processing. FP16.